

**Definition:**

1. i) Crystalline solid    ii) Lattice    iii) Basis    iv) Crystal    v) Unit Cell\*\*\*
2. Miller Indices?\*\*\*
3. Coordination Number?\*\*\*
4. Postulate Of the special theory of relativity\*\*\*
5. What is X-ray? Properties of Xray?\*\*\*
6. Nuclear Fission & Fusion?\*\*\*
7. Half life & Mean life?\*\*\*
8. State the law of disintegration?\*
9. What is radioactivity?
10. i) Photoelectric effect    ii) Compton effect
11. Draw the binding energy curve?

**Expression****Solid State Physics**

1. Packing Fraction (SC, BCC, FCC). \*\*\*
2. Show that in a crystal of cubic structure, the distance between the planes with miller indices h, k, l is equal to  $d = \frac{a}{\sqrt{h^2+k^2+l^2}}$ . \*\*\*
3. Miller Indices (001,110,101,111,100,010,011) \*\*\*
4. Explain The co-ordination number of **NaCl** \*\*\*

**MATH PROBLEM:** 19.3-19.5, 19.15-19.18 =>Page: 859

**Crystal Defect**

1. Distinction between metal, insulator and semiconductor in terms of energy band. \*\*\*
2. How many types of crystal bonding? Explain all of the crystal bonding. \*
3. How many types of crystal defects in solid? Explain all of the crystal defects.

**Relativity**

1. Time dilation "A moving clock always appears to go slow" \*\*\*
2. Einstein Mass-Energy Relation? \*\*\*
3. Variation of Mass with velocity?

**MATH PROBLEM:** (11.12\*\*\*, 11.13, 11.14) (11.21, 11.22, 11.25, 11.26, 11.32, 11.34) =>Page: 859

**Modern Physics**

1. Calculate the following properties of  $^{64}_{28}\text{Ni}$ ,  $^{226}_{88}\text{Ra}$ ,  $^{12}_6\text{C}$ ,  $^{238}_{92}\text{U}$  \*\*\*
  - i. Nuclear Mass
  - ii. Nuclear Size
  - iii. Nuclear Density
  - iv. Nuclear Charge
  - v. Nuclear Mass defect
  - vi. Nuclear Binding Energy
  - vii. Nuclear Binding Energy per nucleon.
2. Calculation of the radii of Bohr Model.
3. Bragg's law and its expression? \*

**MATH PROBLEM:** (13.6, 13.7, 13.8) =>Page: 501

**Radioactivity**

1. Half-life + expression of half-life " $T_{1/2}$  inversely proportional to the radioactive constant". \*\*\*
2. Mean life + expression of mean life \*\*\*
3. Expression of Disintegration law.

**MATH PROBLEM:** (16.1, 16.6, 16.2) =>Page: 707